

A DIGITAL WALKIE TALKIE ARCHITECTURE UTILIZING GFSK

Project Abstract

Institution: National Institute of Technology Goa

Department: Electronics and Communication Engineering

Course: EC256: Electronics and Communication Laboratory

Team Members: Kartikey Tiwari, Karun Raikar, Kathan Kuchekar, Palak, Mayuresh Nalavade

ABSTRACT

This project details the design and implementation of a digital half-duplex communication system functioning as a wireless walkie-talkie within the 2.4 GHz Industrial, Scientific, and Medical (ISM) band. The architectural workflow initiates with the acquisition of an analog acoustic signal via an electret microphone, which undergoes active pre-amplification using an LM358 operational amplifier. This conditions the microvolt-level acoustic signals to an optimal dynamic range before being digitized by an ATmega328P microcontroller-based Analog-to-Digital Converter (ADC).

The discrete data stream is subsequently transmitted via an NRF24L01 radio frequency transceiver module employing Gaussian Frequency Shift Keying (GFSK) modulation. This method ensures robust, low-interference wireless transport by smoothing digital pulses through a Gaussian filter before frequency modulating the carrier wave. At the receiving node, the incident payload is demodulated, error-checked, and synthesized back into an analog waveform utilizing a high-frequency Pulse Width Modulated (PWM) Digital-to-Analog Converter (DAC) architecture executed by the receiving microcontroller.

To achieve high audio fidelity, a passive RC low-pass filter is implemented to attenuate high-frequency quantization and switching artifacts from the raw PWM signal. Finally, power amplification is provided via an LM386 module to drive an 8-Ohm electrodynamic speaker. To validate the hardware implementation, the underlying mathematical principles of the GFSK modulation and demodulation processes, alongside the theories of sampling and quantization, were rigorously modeled and verified using MATLAB. Ultimately, the system provides a robust framework for understanding discrete signal processing and digital radio frequency transmission tailored for localized, off-grid environments independent of commercial network infrastructure.